

GONUGUNTA VENKATA SAI MOTHISH

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Education

2024 - present	Indian Institute Of Science, Bengaluru PhD Computational Data Sciences - Google PhD Fellow 2025 (GPA: 9.10/10.0)
2022 - 2024	Indian Institute Of Science, Bengaluru M.Tech Robotics and Autonomous Systems (GPA: 9.32/10.0)
2018 - 2022	Presidency University, Bengaluru B.Tech in Mechanical Engineering (Gold Medalist) (GPA: 9.17/10.0)
2016 - 2018	Narayana Junior College, Nellore Class 12th Board of Intermediate Education, Andhra Pradesh (Percentage: 95.8 %)
2016	Don Bosco English Medium High School, Nellore Class 10th Board of Secondary Education, Andhra Pradesh (GPA: 9.7/10.0)

Areas of Interests

- Vision Language Models
- Computer Vision
- Optimization
- Reinforcement Learning
- Robotics
- Representation Learning

Relevant Coursework @ IISC

- Digital Image Processing
- Stochastic Models and Applications
- Machine Learning for Signal Processing
- Data Science for Smart City Applications
- Theory and Applications of Bayesian Learning
- Reinforcement Learning
- Human-Computer Interaction
- Robotic Perception

Skills

Frameworks	Pytorch, Tensorflow, Scikit-learn, Pandas, OpenCV, Keras
Technologies	Intel oneAPI, OpenAI gym, Nvidia Issac gym
Languages	Python, C, Matlab

Research Publications (First Authored)

Conditional Diffusion Model with Nonlinear Data Transformation for Time Series Forecasting.
Accepted at **ICML 2025**

DNOD: Deformable Neural Operators for Object Detection in SAR Images.
Accepted at **ICLR 2026**

BiRoDiff: Diffusion policies for bipedal robot locomotion on unseen terrains.
Published at **IEEE - ICC 2024**

Stoch BiRo: Design and Control of a low cost bipedal robot.
Published at **IEEE - ICCAR 2024**

RAF-Diff: Rational Adaptive Functional Diffusion for Time Series Analysis.
Under Review at **NeurIPS 2026**

Research Publications (Co-Authored)

Inverse Reinforcement Learning based Multimodal Target Prediction for Rapid HRI.

Published at ACM - IUI 2024

GLoW: Gabor Localized Latent Wavelet Operator for Solving PDEs on Complex Geometries.

Under Review at NeurIPS 2026

DAWN-Cast: Latent Dynamic Adaptive Wavelet Network for Precipitation Nowcasting.

Under Review at NeurIPS 2026

Projects

[Predicting Missing Edges in Social Networks Using Machine Learning](#) | *Sklearn, Pytorch* Oct 2023

- Objective: Predict missing edges in social networks via machine learning on **Twitter data**.
- Approach: Utilized **NetworkX** for data prep, explored various feature sets, and employed logistic regression and **Adaboost** models. Experimented with sampling strategies and additional features.
- Outcomes: Logistic regression achieved 0.89 AUC, improved to 0.92 with Adaboost. TorchBigGraph yielded 0.9086 AUC on Kaggle, showcasing efforts to enhance accuracy.
- Conclusion: Successfully tackled link prediction challenges, emphasizing iterative feature exploration and model refinement for accurate results.

[Bayesian Imputation for Missing Sensor Data in IoT Devices](#) | *Keras, Pytorch, LSTM* Mar 2023

- Aim: Predict missing values in **IoT sensor data (temperature and humidity)** using Bayesian methods.
- Phases: Data analysis, pre-processing, and transitioning from Frequentist to Bayesian approaches.
- Bayesian Models: **Bayesian Ridge Regression, Gaussian Process Regression** used for accurate imputation after the transition from Frequentist approach (LSTM).

[Automated Driver Assistance System](#) | *Pytorch, TensorFlow, CNN* Jun 2023

- **Object detection (Yolo)** and **Lane Detection** (UltraFast Lane detector) module was developed using pytorch framework. While the **Depth Estimation (MiDaS)** was developed using TensorFlow framework.
- The inference time of the model was optimized using the **Intel oneAPI Deep Neural Network Library** on the intel's developer's cloud.
- A performance speedup of approximately 4.5x was achieved using the oneAPI libraries.

[Estimation of Vehicle Speed using Computer Vision](#) | *Pytorch, OpenCV, CNN* Apr 2023

- This was developed as a part of coursework for Robotic Perception course.
- **PWC-NET** a CNN-based approach was used to estimate the **Optical flow** from the successive video frames.
- Yolo V5 was used for **object detection** in the given frame.
- Calculated the instantaneous velocity by predicting object's movement with optical flow vectors

- Relevant Concepts : **Reinforcement Learning, Deep Learning**
- This my master's project at Stochastic Robotics lab
- STAGE 1: Design and Implement a learning-based controller on a Bipedal walking robot.
- STAGE 2 : Achieving the stable walking through **proprioception** using **Reinforcement learning** blended with physics-based techniques.
- Please refer to the hyperlink for recent progress.

Awards and Achievements

Google PhD Fellow 2025 for Machine Perception

University Gold Medal in Bachelor's Degree

AI and Robotics Technology Park Fellowship 2023

2nd place in Intel OneAPI Hackathon

Jun 2023

- Worked with **Intel AI Analytics Toolkits** and Intel-optimized frameworks such as TensorFlow and PyTorch.
- **Real-time object detection, lane detection and depth estimation** for autonomous vehicles.

Finalist in Falling Walls Lab India

Mar 2019

- Organized by the **German Centre for Research and Innovation (DWIH)** and **DAAD**.
- Theme: Breaking the wall of **irrigation challenges with ML and IoT**.
- Automated plant watering system developed using machine learning (chili crop), guided by Prof. Raghavendra M. Deshpande and Prof. Shashidhar V.